

DATA REPRESENTATION – TYPES OF GRAPHS

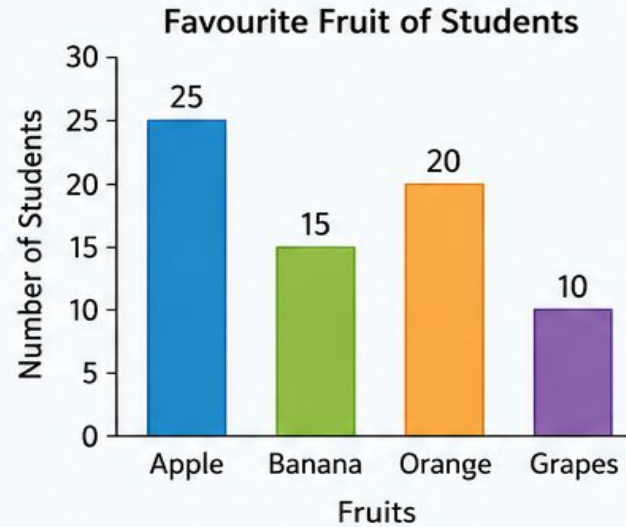
1. BAR CHARTS

Definition:

A bar chart uses rectangular bars of equal width to compare categories. The height of each bar shows the value or frequency of that category.

Example:

Favourite Fruit of Students



2. PICTOGRAMS

Definition:

A pictogram uses pictures or symbols to show data. Each picture stands for a fixed number of items.

Example:

Number of Books Read by Students

Number of Books	Number of Students
0 – 2	 
3 – 5	  
6 – 8	   
9 and above	 

 = 5 students

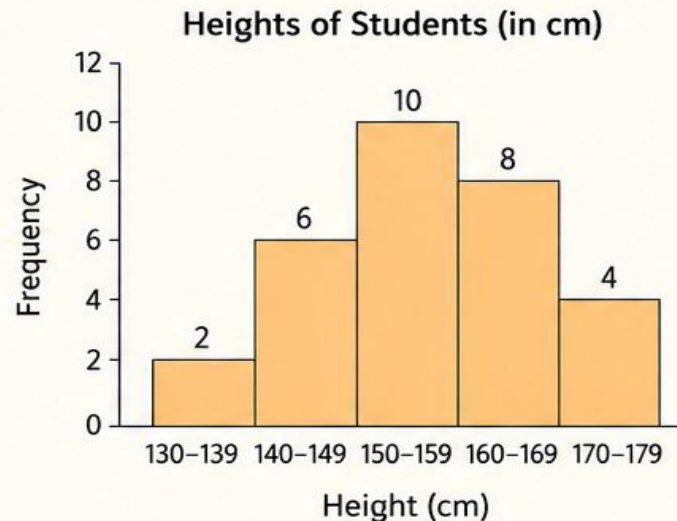
3. HISTOGRAM

Definition:

A histogram is used to show the distribution of continuous data. The bars are joined together. The width of each bar represents a class interval, and the height shows the frequency.

Example:

Heights of Students (in cm)



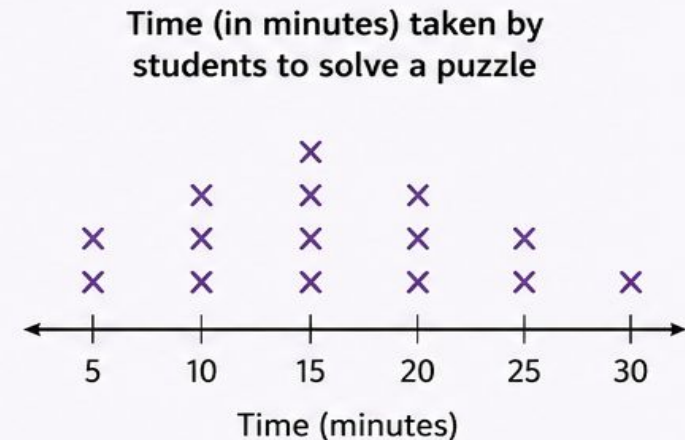
4. LINE PLOTS

Definition:

A line plot (or dot plot) shows data on a number line. Each "X" represents one data value. It is useful for small data sets.

Example:

Time (in minutes) taken by students to solve a puzzle



Key Idea:

These graphs help us organise data and understand it easily.

HOW TO DRAW X-AXIS AND Y-AXIS ON A GRAPH AND LABELING

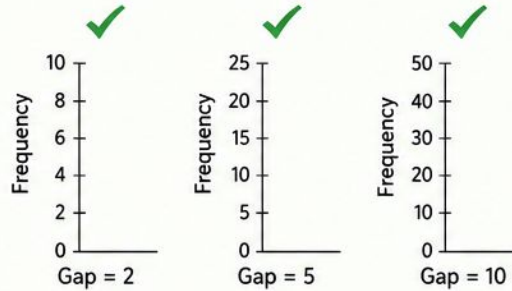
Follow these steps to draw any graph correctly.

STEP-BY-STEP INSTRUCTIONS

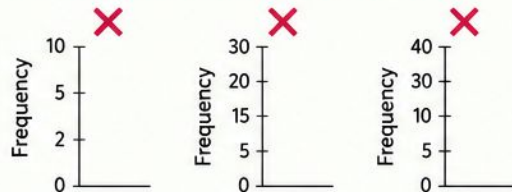
- 1 Draw the axes**
 - Draw a horizontal line. This is the X-axis.
 - Draw a vertical line. This is the Y-axis.
 - Make sure the two axes meet at 0.
- 2 Label the X-axis**
 - Write the name of the data or category below the X-axis.
Example: Favourite Fruits, Time (minutes), Height (cm)
- 3 Label the Y-axis**
 - Write what is being measured on the Y-axis.
Example: Number of Students, Frequency, Number of Books
- 4 Choose a suitable scale for the Y-axis**
 - The gaps between the numbers must be equal.
- 5 Choose the scale according to the data**
 - Find the largest frequency/value.
 - Choose a scale that reaches slightly above the largest value.
 - Keep the same gap throughout the Y-axis.
- 6 Add a title**
 - Write a clear and simple title at the top.

SCALE / GAP BETWEEN FREQUENCY

The numbers on the Y-axis must be in equal gaps.
Examples of correct scales:

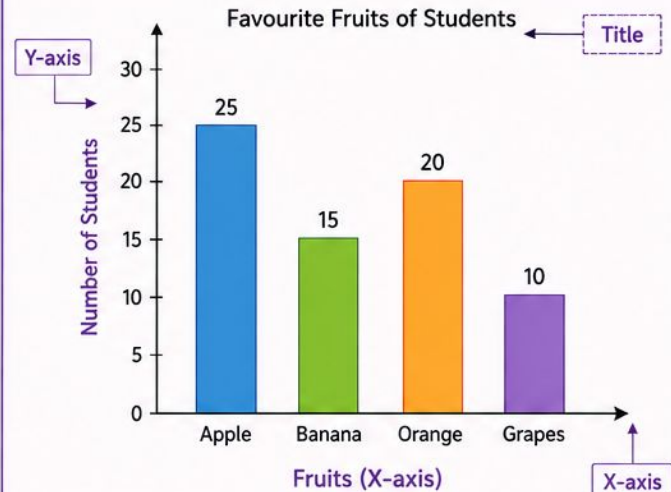


Examples of incorrect scales (unequal gaps):



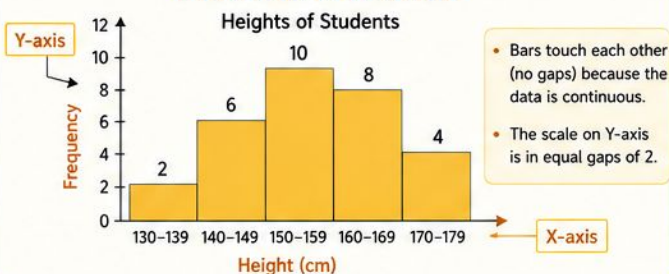
Remember: Always use equal intervals.

EXAMPLE OF A BAR CHART



- X-axis shows the categories (Fruits).
- Y-axis shows the number of students.
- The scale on Y-axis is in equal gaps of 5.

EXAMPLE OF A HISTOGRAM

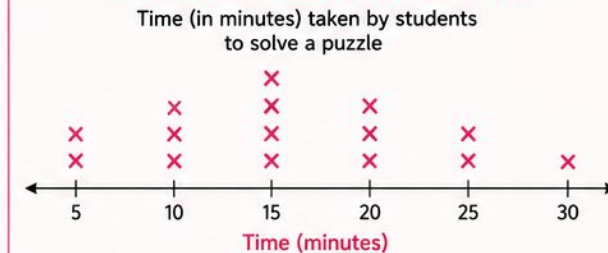


EXAMPLE OF A PICTOGRAM (PICTOGRAPH)

Number of Books Read (in a week)	Number of Students
0 - 2	 
3 - 5	  
6 - 8	    
9 and above	 

 = 5 students

EXAMPLE OF A LINE PLOT (DOT PLOT)



★ QUICK REMINDER : X-axis = What we are studying | Y-axis = How many / Frequency | Equal gaps = Equal intervals | Title = What the graph is about ★